1 Methodology

The GeoZK-Miner workflow governs the lifecycle of conflict minerals through a three-layer architecture. The logic is defined in Algorithm 1.

Algorithm 1 GeoZK-Miner Workflow for Conflict Mineral Provenance and Reacquisition Governance

Input:
  Stakeholder set S = {mine, logistics, smelter, manufacturer, auditor}
  Mineral batch data B = {batchID, mineralType, extractionDate, certificationHash, metadataURI}
  Private location witness L = (x, y, r)
  Approved region boundary R = {minX, maxX, minY, maxY}
Output: Verified mineral batch record, lifecycle history, and reacquisition decision

Step 1: System Initialization
Deploy stakeholder registry, region proof verifier, and GeoZK-Miner core smart contracts.
Register stakeholders S with role-based permissions.
Step 2: Region Approval
Auditor defines approved mining region R.
Compute regionHash = Poseidon(minX, maxX, minY, maxY).
Store regionHash as an approved region on-chain.
Step 3: Mineral Batch Creation
Mine site prepares batch data B.
Compute geoCommitment = Poseidon(x, y, r).
Submit {batchID, mineralType, certificationHash, geoCommitment, regionHash, metadataURI} to contract.
Contract sets status to CREATED and emits BatchCreated event.
Step 4: Region-Based ZKP Generation
Mine site generates ZKP piR using private witness (x, y, r).
Ensure: geoCommitment matches witness AND (x, y) is within R.
Step 5: On-Chain Proof Verification
if Verifier contract accepts piR AND regionHash is approved **then**
    Update status to ZKPVerified; Emit ZKPVerified event.
else
    Reject verification; Mark for manual audit.
end if
Step 6: Lifecycle and Custody Progression
Stakeholders (Logistics, Smelter, Manufacturer) update status as mineral moves.
Each update is stored as an immutable event log.
Step 7: Reacquisition Governance
if Provenance concern raised by Smelter or Manufacturer **then**
    Record ReacquisitionRequested.
    Auditor reviews evidence and selects: APPROVED, REJECTED, or AuditRequired.
    Update final state and emit ReacquisitionResolved.
end if
Step 8: Audit Trail
Return final batch status and full immutable event history.